Check for updates

THERAPEUTIC ADVANCES in
Medical Oncology

Original Research

Real-world comparative effectiveness of apalutamide versus enzalutamide for prostate-specific antigen response in metastatic castration-sensitive prostate cancer

Ther Adv Med Oncol

2026, Vol. 18: 1–13

DOI: 10.1177/17588359261445055

© The Author(s), 2026.
Article reuse guidelines: sagepub.com/journals-permissions

Mehmet A. Bilen, Benjamin H. Lowentritt, Sabree Burbage, Charmi Patel, Carmine Rossi, Frederic Kinkead, Gordon Wong, Dominic Pilon and Neal D. Shore

Abstract

Background: Deep and rapid prostate-specific antigen (PSA) response is an important indicator to consider in treatment decision-making given its predictive value for long-term clinical outcomes in metastatic castration-sensitive prostate cancer (mCSPC).

Objectives: This study compared early PSA90 response (⩾90% reduction in PSA levels) in a large and recent cohort of patients with mCSPC initiating apalutamide or enzalutamide in the United States.

Design: A retrospective longitudinal analysis was conducted.

Methods: Patients with mCSPC were identified from linked claims and electronic medical record data from the United States and categorized into the apalutamide or enzalutamide cohorts based on the first claim/dispensation on or after December 16th, 2019. To adjust for differences observed in patient characteristics between the two cohorts, inverse probability of treatment weighting was used. Weighted Cox proportional hazards models were used to compare PSA90 response (PSA measurement ⩾90% lower than the most recent pretreatment PSA value) by 6 months post treatment initiation between the apalutamide and enzalutamide cohorts.

Results: Among 1296 patients with mCSPC treated with apalutamide and 1111 treated with enzalutamide, pretreatment characteristics were balanced after weighting. PSA90 response by 6 months was achieved by 61.7% of patients in the apalutamide cohort and 55.2% of patients in the enzalutamide cohort (weighted hazard ratio: 1.20 [95% confidence interval: 1.07, 1.36]; $p = 0.003$). Patients in the apalutamide cohort achieved a PSA90 response earlier than those in the enzalutamide cohort (median time to PSA90 response: 3.7 vs 4.9 months, respectively).

Conclusion: In a large, real-world, retrospective analysis, patients with mCSPC treated with apalutamide were associated with a significantly higher rate of early PSA90 response compared to enzalutamide.

Plain language summary

Comparing two treatment options for metastatic castration-sensitive prostate cancer: apalutamide versus enzalutamide

This real-world study compared two medications used to treat men with advanced prostate cancer that has spread but still responds to hormone treatment. These oral medicines, apalutamide and enzalutamide, are taken with hormone therapy to help slow or stop

Correspondence to:
Neal D. Shore
START Carolinas/Carolina Urologic Research Center, 823 82nd Parkway, Myrtle Beach, SC 29572, USA
nshore@auclinics.com

Mehmet A. Bilen
Winship Cancer Institute of Emory University, Atlanta, GA, USA

Benjamin H. Lowentritt
Chesapeake Urology, Towson, MD, USA

Sabree Burbage
Charmi Patel
Johnson & Johnson, Horsham, PA, USA

Carmine Rossi
Frederic Kinkead
Gordon Wong
Dominic Pilon
Analysis Group, Inc., Montréal, QC, Canada

Creative Commons Non Commercial CC BY-NC: This article is distributed under the terms of the Creative Commons Attribution-NonCommercial 4.0 License (https://creativecommons.org/licenses/by-nc/4.0/) which permits non-commercial use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the Sage and Open Access pages (https://us.sagepub.com/en-us/nam/open-access-at-sage).

cancer growth. Physicians often use a blood test called prostate-specific antigen (PSA) to see how well the treatment is working. A large drop in PSA, by 90% or more, is called a PSA90 response and achieving this quicker can be a sign that treatment is working well and may help patients live longer. We reviewed insurance and medical records from more than 2,400 men with metastatic castration-sensitive prostate cancer in the United States who started either apalutamide or enzalutamide between 2019 and 2023 and compared how many patients reached a PSA90 response within six months of starting treatment. A larger percentage of patients treated with apalutamide (62%) achieved a PSA90 response by six months post-treatment than patients treated with enzalutamide (55%). Patients treated with apalutamide reached PSA90, on average, one month faster than those patients treated with enzalutamide. The findings from this study suggest an important clinical benefit of treatment with apalutamide in terms of producing a rapid and deep PSA response among patients with metastatic castration-sensitive prostate cancer.

Keywords: androgen receptor pathway inhibitors, castration-sensitive, comparative effectiveness, hormonal antineoplastic agents, prostate-specific antigen

Received: 12 August 2025; revised manuscript accepted: 8 April 2026.

Introduction

Prostate cancer (PC) is among the most common malignancies in men in the United States (US), with an estimated 313,780 new cases in 2025 and a prevalence of over 3.5 million in 2022.[1,2] The 5-year survival rate for patients with metastasis is 38%,[1] highlighting the poor prognosis and incurable nature of metastatic PC.[3]

Androgen deprivation therapy (ADT) is the backbone of systemic therapy for metastatic PC, and patients who are responsive to ADT are categorized as having metastatic castration-sensitive PC (mCSPC, also referred to as metastatic hormone-sensitive PC).[3,4] In the last decade, clinical trials have demonstrated that the addition of androgen receptor pathway inhibitors (ARPIs) to ADT significantly improves survival outcomes in patients with mCSPC.[5] In particular, the second-generation ARPI apalutamide, in combination with ADT, was associated with significantly longer radiographic progression-free survival (rPFS) (hazard ratio (HR): 0.48 [95% confidence interval (CI): 0.39, 0.60], $p < 0.001$) and overall survival (OS) (HR: 0.67 [95% CI: 0.51, 0.89], $p = 0.005$) compared to placebo at 24 months in the phase III TITAN clinical trial,[5,6] resulting in its Food and Drug Administration approval for mCSPC on September 17, 2019.[7] Another ARPI, enzalutamide, was subsequently approved for the treatment of mCSPC on December 16, 2019,[8] based on its significantly longer rPFS in combination with ADT compared to placebo (HR: 0.39 [95% CI: 0.30, 0.50], $p < 0.001$) in the phase III ARCHES trial.[9]

In post hoc analyses of the TITAN and ARCHES trials, deep and rapid prostate-specific antigen (PSA) responses (e.g., achievement of a ⩾90% reduction in PSA levels (PSA90) or PSA ⩽ 0.2 ng/mL by 3 or 6 months) were recently shown to be associated with improved rPFS and OS.[10,11] Furthermore, a recent real-world study found that among patients treated with ARPIs, those who reached a PSA90 response within 6 months had significantly better OS outcomes than those without an early response. As such, early PSA response should be an important factor to consider in treatment decision-making given its predictive value for long-term clinical outcomes.[12]

In the real world, a recent study of patients with mCSPC in the US found that treatment with apalutamide was associated with a significantly higher likelihood of achieving a PSA90 response by 6 months compared to enzalutamide, along with a faster time to PSA90 response.[13] A small number of other analyses have similarly compared attainment of PSA90 response between patients treated with apalutamide or enzalutamide, with

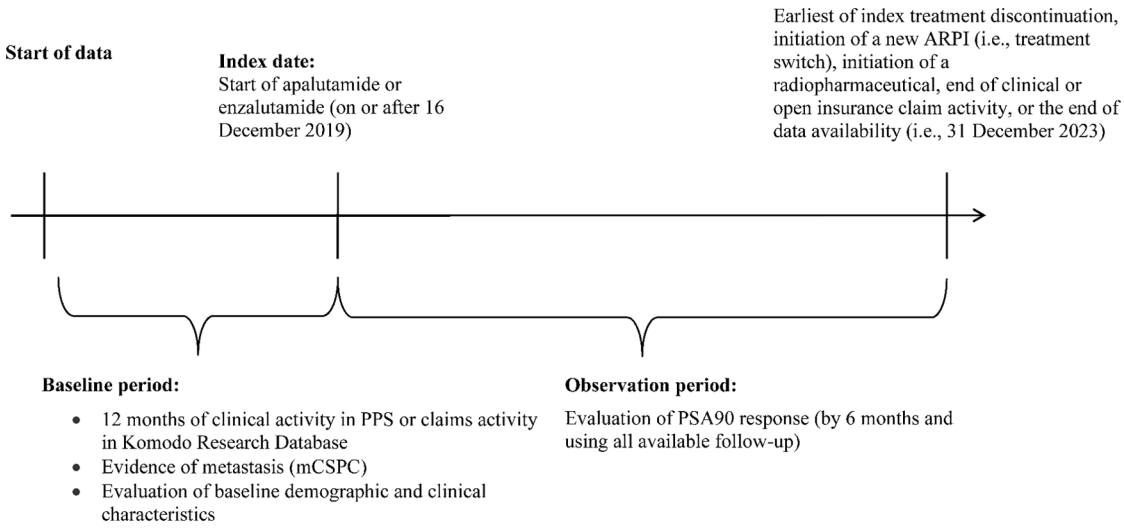


Figure 1. Study design.
ARPI, androgen receptor pathway inhibitor; mCSPC, metastatic castration-sensitive prostate cancer; PPS, Precision Point Specialty; PSA90, ⩾90% reduction in prostate-specific antigen levels.

consistent findings.[14–16] Given the lack of head-to-head trials between apalutamide and enzalutamide nor any consensus recommendations for ARPI selection in the mCSPC setting,[17] a large base of robust real-world data is needed to establish evidence that may help guide treatment decisions in clinical practice. Therefore, this retrospective analysis was conducted to confirm the prior literature by comparing early PSA90 response in a large and recent cohort of patients with mCSPC initiating apalutamide or enzalutamide in the United States.

Materials and methods

Data sources

This analysis used administrative claims data from Komodo Research Database (KRD+, January 1, 2016 to December 31, 2023), linked to clinical electronic medical records (EMRs) from Precision Point Specialty (PPS) Analytics (January 1, 2000–December 31, 2023). Data from KRD+ comprised inpatient and outpatient medical claims with diagnosis and procedure codes, paid pharmacy claims, and information on patient demographics and mortality. The EMR data were collected as part of routine clinical care from >90 private, community-based urology practices across the US and included patient demographic and clinical information (e.g., ARPI dispensations, ADT administrations), metastasis information, assessment of castration resistance, PSA measurements, and prescriptions for advanced PC medications.

This study was considered exempt research under 45 CFR § 46.104(d)(4) as it involved only the secondary use of data that were de-identified in compliance with the Health Insurance Portability and Accountability Act, specifically, 45 CFR § 164.514. The reporting of this study conforms to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement (Supplemental Material).[18]

Study design

A retrospective longitudinal analysis was conducted evaluating real-world effectiveness between apalutamide and enzalutamide (Figure 1). The index date was defined as the date of the first paid pharmacy claim (in the administrative claims) or first dispensation (in the EMR data) of apalutamide or enzalutamide on or following December 16, 2019 (i.e., the Food and Drug Administration approval of enzalutamide, which was the most recent between the two drugs).[7,8] The baseline period comprised the 12-month period prior to the index date. The observation period spanned from the index date until the earliest of index treatment discontinuation (defined by a 90-day treatment gap), initiation of a new ARPI (i.e., treatment switch, excluding first-generation antiandrogens), initiation of a radiopharmaceutical, end of clinical activity in PPS or the end of

open insurance claim activity in the KRD+, or the end of data availability (i.e., December 31, 2023).

Patient selection

Patients were included in the study if they met the following criteria: (1) had ⩾1 paid pharmacy claim or dispensation of an ARPI on or following December 16, 2019, with patients stratified into the apalutamide or enzalutamide cohort if apalutamide or enzalutamide, respectively, was the first (i.e., index) ARPI, (2) did not have paid pharmacy claims, dispensations, or prescriptions (identified in the EMR data) for any ARPI other than the index ARPI on or prior to the index date, (3) had mCSPC on the index date, (4) had ⩾12 months of claims activity or clinical activity (in the EMR data) prior to the index date, (5) were aged ⩾18 years on the index date, (6) had no other prior primary cancer diagnoses during the baseline period, (7) did not use estrogens, immunotherapy, poly (ADP-ribose) polymerase inhibitors, radiopharmaceutical therapy, or etoposide, (8) did not use cabazitaxel or carboplatin after docetaxel any time prior to or on the index date, and (9) had ⩾1 PSA measurement during the 13 weeks prior to and including the index date (Figure 2).

Study measures and outcomes

Patient characteristics evaluated during the baseline period included demographics (age, race, geographic region, payer type) and clinical information (year of index ARPI initiation, time between metastasis and ARPI initiation, time between PC diagnosis and ARPI initiation, metastasis type (i.e., bone, nodal, and visceral), *de novo* mCSPC (i.e., ⩽180 days between the first observed PC diagnosis and metastasis), concurrent and prior use of ADT, prior use of first-generation androgen receptor inhibitor, chemotherapy use, PSA and testosterone levels, Gleason score).

Outcomes evaluated during the observation period included the proportion of patients achieving a PSA90 response by 6 months post index and over the entire observation period, the median time to PSA90 response, and post-index PSA testing patterns. A PSA90 response was defined as a PSA measurement while on the index ARPI that was ⩾90% lower than the most recent baseline PSA value observed within the 13 weeks up to and including the index date.

Statistical analysis

The null hypothesis was that the proportion of patients achieving a PSA90 response by 6 months was the same in the apalutamide and enzalutamide cohorts, while the alternative hypothesis was that this proportion was not the same between the two cohorts.

To adjust for differences observed in the baseline characteristics of patients in the apalutamide and enzalutamide cohorts, inverse probability of treatment weighting (IPTW) using propensity scores (PS).[19] The PS was generated using probability estimates from a logistic regression model with the following predictors: age (continuous), race, geographic region, payer, year of index date (categorical), time between PC diagnosis and index date (continuous), time between metastasis and index date (continuous and categorical), metastasis type (bone, nodal, visceral, metastasis in multiple sites), *de novo* mCSPC, Quan–Charlson comorbidity index (continuous), concurrent ADT use, first-generation ARPI use, chemotherapy use, most recent PSA level (categorical), and earliest Gleason score (categorical). Patients in the apalutamide cohort (treated cohort) were assigned a weight of 1/(PS) and those in the enzalutamide cohort (control cohort) were assigned a weight of 1/(1-PS). Normalized IPTWs were truncated at the 95th percentiles to prevent undue influence from extreme weights. The number of patients in each cohort remained unchanged following weighting. Patient baseline characteristics were compared between the two unweighted and weighted cohorts using standardized differences, with a standardized difference $<10\%$ indicating balance.[20]

The proportion of patients achieving a PSA90 response by 6 months post index and over the entire observation period was estimated using weighted Kaplan–Meier analysis. The association between apalutamide or enzalutamide treatment and achieving a PSA90 response was evaluated using weighted Cox proportional hazards models to generate HRs and their 95% CIs.

Results

Patient characteristics

The study included 1296 patients with mCSPC in the apalutamide cohort and 1111 patients with mCSPC in the enzalutamide cohort. After applying IPTW, standardized differences were generally

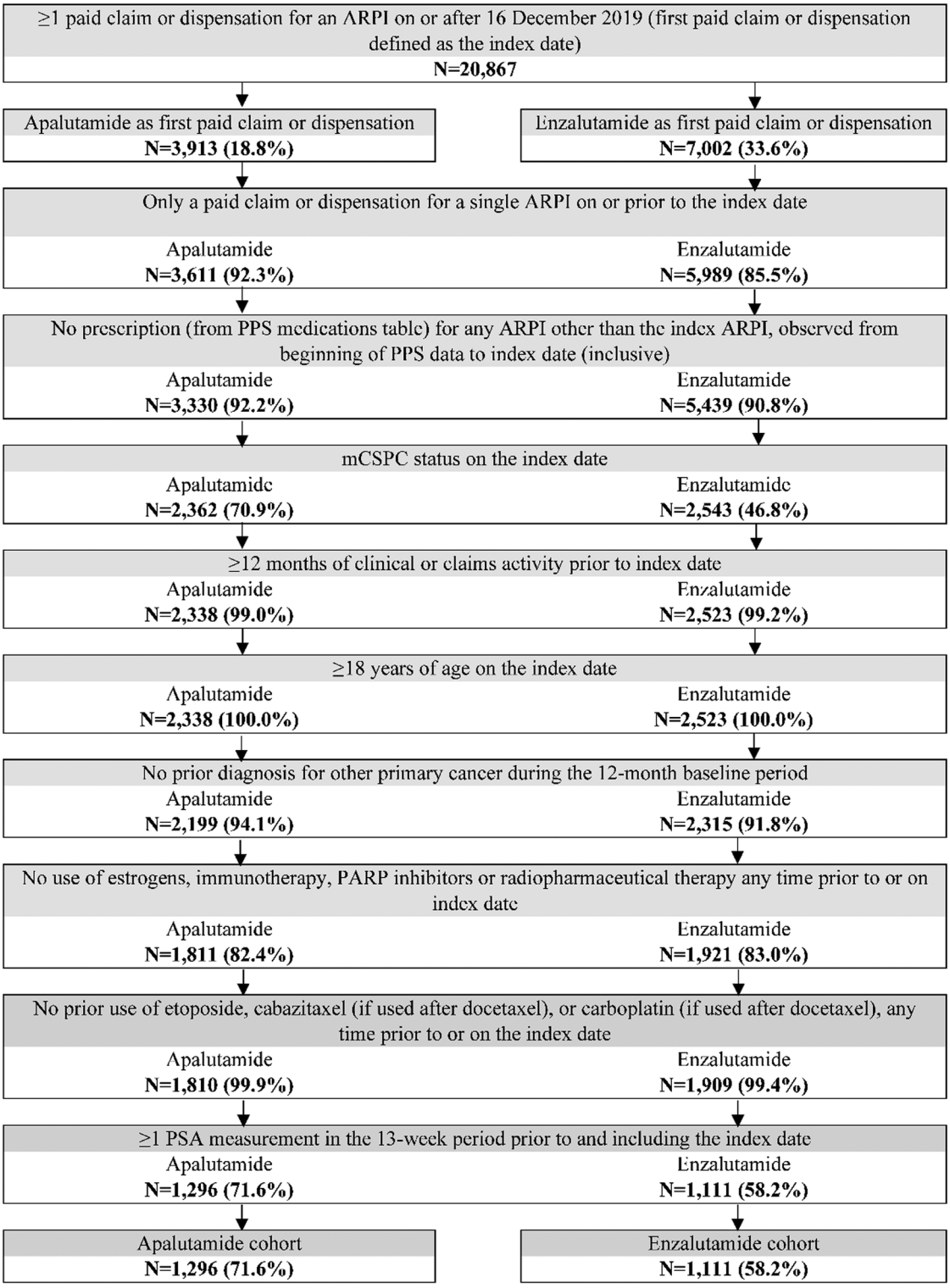


Figure 2. Patient flowchart.
ARPI, androgen receptor pathway inhibitor; mCSPC, metastatic castration-sensitive prostate cancer; PARP, poly ADP ribose polymerase; PPS, Precision Point Specialty; PSA, prostate-specific antigen.

<10% for baseline characteristics, indicating that the weighted cohorts were well-balanced (Table 1). The mean (median) age was 73.4 (74.0) years in the apalutamide cohort and 73.4 (73.0) years in the enzalutamide cohort, and 21.6% and 21.7%, respectively, were Black/African American. More than half of the patients were from the US South (apalutamide: 55.1%, enzalutamide: 54.5%), and Medicare was the most common type of insurance coverage (apalutamide: 79.9%, enzalutamide: 80.1%). Nearly a quarter of patients in both cohorts initiated their index ARPI in 2023 (apalutamide: 24.7%, enzalutamide: 24.1%)

Patients had a mean (median) time between metastasis and ARPI initiation of 8.9 (2.5) months in the apalutamide cohort and 9.0 (2.4) months in the enzalutamide cohort. Approximately 53.6% of patients in the apalutamide cohort and 53.7% of patients in the enzalutamide cohort had *de novo* mCSPC. During the baseline period, the mean (median) PSA level was 27.4 (3.6) ng/mL in the apalutamide cohort and 27.0 (3.5) ng/mL in the enzalutamide cohort.

Post-index PSA90 response

The mean (median) on-treatment period was 11.1 (7.5) months in the apalutamide cohort and 9.9 (6.9) months in the enzalutamide cohort. Within 6 months of the index date, a PSA90 response was achieved by 61.7% of patients in the apalutamide cohort and 55.2% of patients in the enzalutamide cohort (weighted HR: 1.20 [95% CI: 1.07, 1.36], $p=0.003$, Figure 3). Over the entire on-treatment period, patients in the apalutamide cohort had significantly higher PSA90 response rates than those in the enzalutamide cohort (weighted HR: 1.20 [95% CI: 1.07, 1.34], $p=0.002$). Patients in the apalutamide cohort achieved a PSA90 response earlier than those in the enzalutamide cohort (median time to PSA90 response: 3.7 vs 4.9 months, respectively).

Post-index PSA testing patterns

A total of 81.0% of patients in the apalutamide cohort and 77.0% of patients in the enzalutamide cohort had ⩾1 PSA measurement during the observation period, with 79.2% of patients in the apalutamide cohort and 75.2% of patients in the enzalutamide cohort having ⩾1 PSA measurement by 6 months post index (Table 2). Overall, the mean (median) number of PSA tests per year was similar between patients in the apalutamide (4.1 (3.7)) and enzalutamide (3.8 (3.5)) cohorts.

Discussion

In this real-world analysis, administrative claims data linked to clinical EMRs were leveraged to identify a large and recent cohort of patients with mCSPC initiating apalutamide or enzalutamide in US clinical practice between 2019 and 2023. The use of the linked data allowed for the selection of patients with mCSPC in the absence of a mCSPC-specific diagnosis code in the claims, based on proxy indicators available in EMRs like PSA levels. The findings indicated that within 6 months of initiating treatment, patients initiated with apalutamide were associated with a 20% higher rate of PSA90 response compared to patients who initiated enzalutamide. Moreover, this response was achieved more than 1 month earlier with apalutamide relative to enzalutamide.

To our knowledge, this retrospective analysis compares PSA90 response in the largest and most recent sample of patients with mCSPC initiating apalutamide or enzalutamide in clinical practice, with over 2400 patients. As such, it represents a significant contribution to the growing body of real-world evidence demonstrating favorable PSA90 outcomes with apalutamide relative to enzalutamide in the treatment of mCSPC.[13–16] In a prior analysis using the same linked claims and EMR data but with nearly 700 fewer patients, those initiating apalutamide had a 21% greater rate of PSA90 response by 6 months compared to those initiating enzalutamide (62.5% vs 53.8%, weighted HR: 1.21 [95% CI: 1.05, 1.40], $p=0.008$), which was confirmed by the current study findings.[13] Similarly, median time to PSA90 response was shorter with apalutamide (3.7 months) relative to enzalutamide (5.1 months) in the prior analysis. Maughan et al. conducted a separate study using EMRs from US oncology clinics to compare apalutamide and other ARPIs among patients with mCSPC.[16] In their comparison of apalutamide and enzalutamide, the median time to PSA90 response was 3.08 and 4.82 months, respectively, with apalutamide being associated with a higher likelihood of achieving a PSA90 response at any time compared to enzalutamide (adjusted HR: 1.5 [95% CI: 1.2, 1.8], $p<0.001$). These findings are aligned with the present study results and highlight the consistent clinical benefit associated with apalutamide for the treatment of mCSPC in real-world clinical practice.

Table 1. Baseline characteristics.

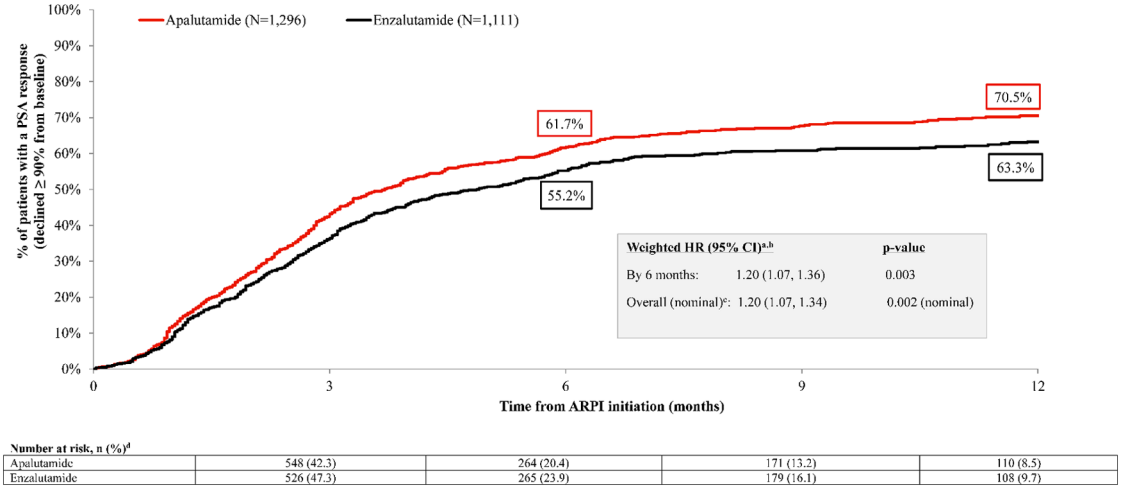
Characteristic	Nonweighted population			Weighted population[a]		
	Apalutamide	Enzalutamide	Standardized difference, %	Apalutamide	Enzalutamide	Standardized difference, %
	N = 1296	*N* = 1111		*N* = 1296	*N*=1111	
Age, mean ± SD [median]	73.3 ± 9.0 [74.0]	73.6 ± 9.0 [73.0]	2.7	73.4 ± 9.0 [74.0]	73.4 ± 9.0 [73.0]	0.3
Race, *n* (%)						
White	815 (62.9)	654 (58.9)	8.2	800 (61.8)	681 (61.3)	1.0
Black or African American	274 (21.1)	255 (23.0)	4.4	280 (21.6)	241 (21.7)	0.3
Hispanic or Latino	87 (6.7)	81 (7.3)	2.3	89 (6.9)	78 (7.0)	0.3
Other	42 (3.2)	55 (5.0)	8.6	50 (3.8)	45 (4.1)	1.2
Unknown	78 (6.0)	66 (5.9)	0.3	77 (5.9)	66 (5.9)	0.1
Geographic region, *n* (%)						
South	736 (56.8)	585 (52.7)	8.3	714 (55.1)	606 (54.5)	1.1
Midwest	293 (22.6)	276 (24.8)	5.3	303 (23.4)	263 (23.7)	0.8
Northeast	140 (10.8)	147 (13.2)	7.5	151 (11.7)	132 (11.8)	0.5
West	127 (9.8)	103 (9.3)	1.8	128 (9.9)	110 (9.9)	0.2
Payer type, *n* (%)						
Medicare	1025 (79.1)	894 (80.5)	3.4	1036 (79.9)	890 (80.1)	0.5
Commercial	230 (17.7)	187 (16.8)	2.4	222 (17.2)	191 (17.2)	0.0
Medicaid	38 (2.9)	28 (2.5)	2.5	35 (2.7)	29 (2.6)	0.7
Unknown	3 (0.2)	2 (0.2)	0.0	3 (0.2)	2 (0.1)	0.0
Index year, *n* (%)						
2019–2020	245 (18.9)	285 (25.7)	16.3	275 (21.2)	243 (21.8)	1.5
2021	324 (25.0)	332 (29.9)	11.0	352 (27.2)	304 (27.4)	0.5
2022	367 (28.3)	272 (24.5)	8.7	349 (26.9)	296 (26.6)	0.6
2023	360 (27.8)	222 (20.0)	18.4	320 (24.7)	268 (24.1)	1.3
Time between metastasis and index date, months, mean ± SD [median]	8.8 ± 17.9 [2.4]	9.3 ± 16.4 [2.7]	3.2	8.9 ± 17.4 [2.5]	9.0 ± 16.8 [2.4]	0.3
Time between PC diagnosis and index date, months, mean ± SD [median]	40.9 ± 47.7 [17.2]	39.5 ± 46.3 [17.1]	3.1	40.1 ± 47.0 [16.8]	40.1 ± 47.2 [17.1]	0.1
Metastasis type,[b] *n* (%)						
Bone	883 (68.1)	791 (71.2)	6.7	900 (69.5)	776 (69.8)	0.8
Nodal	693 (53.5)	535 (48.2)	10.7	665 (51.3)	564 (50.8)	1.0
Visceral	217 (16.7)	208 (18.7)	5.2	226 (17.4)	196 (17.7)	0.7
Metastasis in multiple sites	386 (29.8)	298 (26.8)	6.6	370 (28.6)	315 (28.4)	0.4

(Continued)

Table 1. (Continued)

Characteristic	Nonweighted population			Weighted population[a]		
	Apalutamide	Enzalutamide	Standardized difference, %	Apalutamide	Enzalutamide	Standardized difference, %
	N = 1296	*N* = 1111		*N* = 1296	*N*=1111	
Quan-CCI, mean ± SD [median]	8.6 ± 2.9 [8.0]	8.7 ± 3.1 [9.0]	3.3	8.6 ± 2.9 [9.0]	8.6 ± 3.1 [9.0]	0.0
De novo mCSPC,[c] *n* (%)	688 (53.1)	601 (54.1)	2.0	695 (53.6)	597 (53.7)	0.2
Concurrent use of ADT with index ARPI,[d] *n* (%)	1208 (93.2)	1010 (90.9)	8.5	1204 (92.9)	1018 (91.6)	4.9
Prior use of first-generation ARPI,[e] *n* (%)	189 (14.6)	246 (22.1)	19.6	228 (17.6)	204 (18.4)	2.1
Prior use of chemotherapy,[f] *n* (%)	17 (1.3)	29 (2.6)	9.4	22 (1.7)	22 (1.9)	1.6
Most recent PSA level,[g] ng/ml, mean ± SD [median]	26.1 ± 57.1 [3.5]	28.4 ± 59.6 [3.8]	3.9	27.4 ± 58.3 [3.6]	27.0 ± 57.9 [3.5]	0.7
⩽0.2, *n* (%)	271 (20.9)	206 (18.5)	6.0	261 (20.1)	221 (19.9)	0.7
>0.2 to ⩽2, *n* (%)	263 (20.3)	248 (22.3)	5.0	271 (20.9)	236 (21.3)	0.8
>2 to ⩽5, *n* (%)	180 (13.9)	136 (12.2)	4.9	170 (13.1)	143 (12.9)	0.7
>5 to ⩽10, *n* (%)	152 (11.7)	105 (9.5)	7.4	141 (10.9)	118 (10.6)	0.9
>10, *n* (%)	430 (33.2)	416 (37.4)	8.9	452 (34.9)	393 (35.3)	0.9
Initial Gleason score,[h] *n* (%)						
⩽6	88 (6.8)	64 (5.8)	4.3	83 (6.4)	71 (6.4)	0.0
7	267 (20.6)	191 (17.2)	8.7	249 (19.2)	209 (18.8)	1.0
8	182 (14.0)	172 (15.5)	4.1	193 (14.9)	165 (14.9)	0.0
9	291 (22.5)	243 (21.9)	1.4	287 (22.1)	247 (22.2)	0.2
10	39 (3.0)	48 (4.3)	7.0	46 (3.5)	41 (3.7)	0.9
Unknown	429 (33.1)	393 (35.4)	4.8	438 (33.8)	377 (34.0)	0.3

[a]Of note, the number of patients reported in this weighted population represents the sum of weights for the corresponding nonweighted patients, rounded to the nearest integer. The proportions displayed were calculated before the rounding and may be slightly different than if they were calculated based on rounded numbers.
[b]Types of metastases were defined at any time prior to (and including) the index date. Types of metastases were not mutually exclusive.
[c]*De novo* mCSPC was defined as ⩽180 days between first observed PC diagnosis and date of metastasis.
[d]Concurrent use was defined as having a record for an ADT agent from 180 days before or after the index date.
[e]Prior use of first-generation ARPI was defined as any prescription for bicalutamide, nilutamide, or flutamide in the 12 months preceding the index date.
[f]Prior chemotherapy use was defined as any administration in the 12 months preceding the index date.
[g]Most recent PSA level was assessed within 13 weeks prior to or on the index date.
[h]Gleason score was evaluated at any time prior to and including the index date.
ADT, androgen deprivation therapy; ARPI, androgen receptor pathway inhibitor; CCI, Charlson comorbidity index; mCSPC, metastatic castration-sensitive prostate cancer; PC, prostate cancer; PSA, prostate-specific antigen; SD, standard deviation.



Number at risk, n (%)[d]

Apalutamide	548 (42.3)	264 (20.4)	171 (13.2)	110 (8.5)
Enzalutamide	526 (47.3)	265 (23.9)	179 (16.1)	108 (9.7)

Figure 3. Weighted Kaplan–Meier rates for PSA90. [a]Propensity scores were generated using probability estimates from a logistic regression model using the following predictors: age (continuous), race, geographic region, payer, year of index date, time between metastasis and index date (continuous and categorical), time between PC diagnosis and index date (continuous), *de novo* mCSPC, previous ADT use overlapping with index date, first-generation antiandrogen use, chemotherapy use, types of metastases (bone, nodal, visceral, and metastasis in multiple sites), Quan–Charlson Comorbidity Index (continuous), most recent PSA level (categorical), and earliest Gleason score (categorical). Each patient was attributed an inverse-probability of treatment weight that was defined as follows: 1/(propensity score) for the apalutamide cohort and 1/(1-propensity score) for the enzalutamide cohort. Normalized inverse-probability of treatment weights were truncated at the 95th percentiles. [b]A hazard ratio >1 indicates that the apalutamide cohort had a higher rate of PSA90 compared to the enzalutamide cohort. [c]Given that the primary objective of the study was to compare the PSA90 response between the cohorts by 6 months post index, all other *p*-values were considered nominal. [d]Of note, the number of patients reported in this weighted population represents the sum of weights for the corresponding nonweighted patients, rounded to the nearest integer. The proportions displayed were calculated before the rounding and may be slightly different than if they were calculated based on rounded numbers.
ADT, androgen deprivation therapy; ARPI, androgen receptor pathway inhibitor; CI, confidence interval; HR, hazard ratio; mCSPC, metastatic castration-sensitive prostate cancer; PC, prostate cancer; PSA, prostate-specific antigen, PSA90, ⩾90% reduction in prostate-specific antigen levels.

This study also confirms the robust PSA outcomes reported in the post-hoc analysis of the TITAN clinical trial, where 68% of patients treated with apalutamide achieved a PSA90 response by 6 months.[11] Of note, the recent post-hoc analysis of the ARCHES clinical trial did not evaluate PSA90.[10]

Attainment of early PSA responses like PSA90 has been shown to have predictive value for long-term outcomes in mCSPC.[21–25] With regards to apalutamide, several real-world studies have demonstrated a link between PSA response and OS or rPFS.[21–23,25] For instance, in one analysis of US EMRs, patients initiating apalutamide who achieved a PSA90 response within 3 months had a 90% lower mortality rate compared to those who did not achieve such a response during the study period (adjusted HR: 0.10 [95% CI: 0.04, 0.30], $p < 0.0001$).[23] In another real-world study using administrative claims data from the US, patients with mCSPC initiated on ARPIs who achieved a PSA90 response within 6 months had a 60% lower mortality rate at 36 months (adjusted HR: 0.40 [95% CI: 0.31, 0.50] $p < 0.001$) and 49% lower rate of progression to castration resistance or death at 36 months (adjusted HR: 0.51 [95% CI: 0.43, 0.60] $p < 0.001$).[12] Moreover, an exploratory analysis of the TITAN trial found that early PSA response (⩽0.2 ng/mL) by 6 months was a causal mediator and predictor of treatment effect from apalutamide on OS.[26]

Although the current study did not evaluate survival outcomes in relation to PSA90 response, prior real-world analyses have compared OS among patients with mCSPC treated with apalutamide or enzalutamide.[16,27] In one recent study using the same linked claims and EMR data as the current analysis, Bilen et al. found that treatment with apalutamide was associated with a 23% reduction in the rate of death compared to

Table 2. Follow-up PSA testing.

	Nonweighted population		Weighted population[a]	
	Apalutamide	Enzalutamide	Apalutamide	Enzalutamide
	N = 1296	*N* = 1111	*N* = 1296	*N* = 1111
Patients with ⩾1 PSA test, *n* (%)	1048 (80.9)	849 (76.4)	1049 (81.0)	855 (77.0)
Within 3 months of observation	909 (70.1)	709 (63.8)	911 (70.3)	719 (64.7)
Within 6 months of observation	1025 (79.1)	828 (74.5)	1026 (79.2)	836 (75.2)
Number of follow-up PSA tests per year, mean ± SD [median]	4.1 ± 3.5 [3.7]	3.8 ± 3.6 [3.4]	4.1 ± 3.5 [3.7]	3.8 ± 3.5 [3.5]
Patients with PSA test on average every 3 months, *n* (%)	556 (42.9)	446 (40.1)	555 (42.8)	456 (41.0)
Patients with PSA test on average every 6 months, *n* (%)	980 (75.6)	771 (69.4)	980 (75.7)	782 (70.3)

[a]Of note, the number of patients reported in this weighted population represents the sum of weights for the corresponding nonweighted patients, rounded to the nearest integer. The proportions displayed were calculated before the rounding and may be slightly different than if they were calculated based on rounded numbers.
PSA, prostate-specific antigen; SD, standard deviation.

enzalutamide by 24 months (HR: 0.77 [95% CI: 0.62, 0.96], $p = 0.019$).[27] Of note, the current analysis expanded upon this prior study by evaluating PSA90 response in the subset of patients with available PSA measurements.

In addition to survival, achieving a PSA response is also associated with patient-reported outcomes like quality of life.[28] In a post-hoc analysis of the TITAN trial, patients with mCSPC who achieved a rapid and deep PSA decline at 3 and 6 months of apalutamide treatment experienced prolonged maintenance of health-related quality of life and physical well-being, as well as a longer time to worsening of pain and fatigue symptoms.[28] Importantly, achieving earlier PSA responses seemed to be associated with better quality of life preservation.

Taken together, the current study confirms the significant therapeutic benefit associated with apalutamide compared with enzalutamide in terms of early PSA90 response, with prior literature suggesting additional, long-term clinical implications in the treatment of mCSPC.[21–25] Given the lack of head-to-head clinical trials comparing apalutamide and enzalutamide in the mCSPC setting, the addition of the current real-world comparative findings to the evidence base may be considered to inform treatment decisions among patients initiating ARPIs in clinical practice. Future research may focus on evaluating the impact of attaining a deep and early PSA response on other clinical and patient-reported outcomes and how this differs between apalutamide and enzalutamide.

Limitations

The study findings should be considered in light of several limitations. While the use of linked claims and EMRs allowed for access to a wider range of data, there may have been errors in linking and tokenization between KRD+ and PPS Analytics; however, patients with inconsistent birth and death records between the claims and EMR data were excluded from the analysis to reduce the impact of these errors. Relatedly, omissions, incompleteness, and miscoding in the EMR or claims data may have led to residual and unmeasured confounding between the two cohorts or misclassification of patients as having mCSPC. While the IPTW used to balance patients from both cohorts could only account for measured covariates, the variables included into the model were comprehensive and covered known predictors of PSA90 response in this population. As patients in the apalutamide cohort had a slightly higher rate of PSA measurements than those in the enzalutamide cohort within the first 6 months of treatment, surveillance bias may have resulted in an increased likelihood of

observing a PSA90 response in the former cohort. Additionally, reasons for discontinuing the index ARPI were not available in the claims and EMR data. Lastly, as the study was retrospective and observational in nature, causality could not be established between the index ARPI and outcomes.

Conclusion

In this large, real-world retrospective analysis, patients treated with apalutamide were associated with a statistically significant 20% increase in the rate of achieving an early PSA90 response compared to enzalutamide among patients with mCSPC. This finding is particularly important, given the established link between early PSA response and prolonged survival outcomes in the literature. As such, these results may be used with the existing evidence base to inform treatment decisions among patients initiating ARPIs in clinical practice.

Previous presentations

Part of the material in this manuscript was presented at the American Society of Clinical Oncology Quality Care Symposium (ASCO QCS), held October 10–11, 2025 in Chicago, IL, USA, as a poster presentation.

Declarations

Ethics approval and consent to participate

The study was considered exempt research under 45 CFR § 46.104(d)(4) as it involved only the secondary use of data that were de-identified in compliance with the Health Insurance Portability and Accountability Act, specifically, 45 CFR § 164.514. As such, consent to participate was not required.

Consent for publication

Not applicable.

Author contributions

Mehmet A. Bilen: Conceptualization; Investigation; Methodology; Validation; Writing – original draft; Writing – review & editing.

Benjamin H. Lowentritt: Conceptualization; Investigation; Methodology; Validation; Writing – original draft; Writing – review & editing.

Sabree Burbage: Conceptualization; Investigation; Methodology; Validation; Writing - original draft; Writing – review & editing.

Charmi Patel: Conceptualization; Investigation; Methodology; Validation; Writing – original draft; Writing – review & editing.

Carmine Rossi: Conceptualization; Formal analysis; Investigation; Methodology; Validation; Writing – original draft; Writing – review & editing.

Frederic Kinkead: Conceptualization; Formal analysis; Investigation; Methodology; Software; Validation; Writing – original draft; Writing – review & editing.

Gordon Wong: Formal analysis; Investigation; Methodology; Software; Validation; Writing – original draft; Writing – review & editing.

Dominic Pilon: Conceptualization; Formal analysis; Investigation; Methodology; Project administration; Supervision; Validation; Writing – original draft; Writing – review & editing.

Neal D. Shore: Conceptualization; Investigation; Methodology; Validation; Writing – original draft; Writing – review & editing.

Acknowledgements

Medical writing assistance was provided by professional medical writer, Christine Tam, MWC, an employee of Analysis Group, Inc., a consulting company that has provided paid consulting services to Johnson & Johnson, which funded the development and conduct of this study and manuscript. The authors would like to thank Kruti Joshi, an employee of Johnson & Johnson at the time the study was conducted, for her contributions to the study.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This study was funded by Johnson & Johnson.

Competing interests

M.A.B. is an employee of Winship Cancer Institute of Emory University and has received consulting fees from Johnson & Johnson. B.H.L. is an employee of Chesapeake Urology and has received consulting fees from Johnson & Johnson. S.B. and C.P. are employees and stockholders of Johnson & Johnson. C.R., F.K., G.W., and D.P. are employees of Analysis Group, Inc., a consulting company that has provided paid consulting services to Johnson & Johnson, which funded the development and conduct of this study and

manuscript. N.D.S. is an employee of START Carolinas/Carolina Urologic Research Center and has received consulting fees from Johnson & Johnson.

Availability of data and materials

The data that support the findings of this study are available from Komodo Health Solutions and PPS Analytics. Restrictions apply to the availability of these data, which were used under license for this study.

ORCID iDs

Mehmet A. Bilen https://orcid.org/0000-0003-4003-1103

Carmine Rossi https://orcid.org/0000-0001-5801-6927

Supplemental material

Supplemental material for this article is available online.

References

1. National Cancer Institute, Surveillance, Epidemiology, and End Results Program. Cancer stat facts: prostate cancer, https://seer.cancer.gov/statfacts/html/prost.html (2025, accessed 24 July 2025).
2. Siegel RL, Kratzer TB, Giaquinto AN, et al. Cancer statistics, 2025. *CA Cancer J Clin* 2025; 75: 10–45.
3. Hamid AA, Sayegh N, Tombal B, et al. Metastatic hormone-sensitive prostate cancer: toward an era of adaptive and personalized treatment. *Am Soc Clin Oncol Educ Book* 2023; 43: e390166.
4. Bernal A, Bechler AJ, Mohan K, et al. The current therapeutic landscape for metastatic prostate cancer. *Pharmaceuticals (Basel)* 2024; 17: 20240308.
5. Troy SP, Jakubowski CD and Gartrell BA. Packing the punch: current and emerging treatment strategies in metastatic castration-sensitive prostate cancer. *Curr Urol Rep* 2025; 26: 50.
6. Chi KN, Agarwal N, Bjartell A, et al. Apalutamide for metastatic, castration-sensitive prostate cancer. *N Engl J Med* 2019; 381: 13–24.
7. US Food and Drug Administration. FDA approves apalutamide for metastatic castration-sensitive prostate cancer, https://www.fda.gov/drugs/resources-information-approved-drugs/fda-approves-apalutamide-metastatic-castration-sensitive-prostate-cancer (2019, 7 accessed July 2025).
8. US Food and Drug Administration. FDA approves enzalutamide for metastatic castration-sensitive prostate cancer, https://www.fda.gov/drugs/resources-information-approved-drugs/fda-approves-enzalutamide-metastatic-castration-sensitive-prostate-cancer (2019, accessed July 7, 2025).
9. Armstrong AJ, Szmulewitz RZ, Petrylak DP, et al. ARCHES: a randomized, phase III study of androgen deprivation therapy with enzalutamide or placebo in men with metastatic hormone-sensitive prostate cancer. *J Clin Oncol* 2019; 37: 2974.
10. Azad AA, Petrylak DP, Iguchi T, et al. Enzalutamide and prostate-specific antigen levels in metastatic prostate cancer: a secondary analysis of the ARCHES randomized clinical trial. *JAMA Netw Open* 2025; 8: e258751.
11. Chowdhury S, Bjartell A, Agarwal N, et al. Deep, rapid, and durable prostate-specific antigen decline with apalutamide plus androgen deprivation therapy is associated with longer survival and improved clinical outcomes in TITAN patients with metastatic castration-sensitive prostate cancer. *Ann Oncol* 2023; 34: 477–485.
12. Du S, Rossi C, Khilfeh I, et al. Prostate-specific antigen reduction after androgen receptor pathway inhibitor initiation: real-world comparison of disease progression among patients with metastatic castration-sensitive prostate cancer. *J Health Econ Outcomes Res* 2025; 12: 41.
13. Lowentritt B, Khilfeh I, Pilon D, et al. Real-world comparison of deep prostate-specific antigen response in patients with metastatic castration-sensitive prostate cancer treated with apalutamide or enzalutamide. *Rev Urol* 2024; 23: e29–e40.
14. Lowentritt B, Khilfeh I, Pilon D, et al. Comparison of deep prostate-specific antigen response in patients with de novo metastatic castration-sensitive prostate cancer initiated on apalutamide vs enzalutamide. *Rev Urol* 2025; 24: 1–12.
15. Lowentritt B, Pilon D, Khilfeh I, et al. Attainment of early, deep prostate-specific antigen response in metastatic castration-sensitive prostate cancer: a comparison of patients initiated on apalutamide or enzalutamide. *Urol Oncol* 2023; 41: 253.e251–253.e259.

16. Maughan BL, Liu Y, Mundle S, et al. Survival outcomes of apalutamide as a starting treatment: impact in real-world patients with metastatic hormone sensitive prostate cancer (OASIS). *Prostate Cancer Prostatic Dis* 2024; 1–9. DOI: 10.1038/s41391-024-00929-6.

17. Lowrance W, Dreicer R, Jarrard DF, et al. Updates to advanced prostate cancer: AUA/SUO guideline (2023). *J Urol* 2023; 209: 1082–1090.

18. Equator network. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies, https://www.equator-network.org/reporting-guidelines/strobe/ (2025, accessed 4 March 2026).

19. Austin PC. An introduction to propensity score methods for reducing the effects of confounding in observational studies. *Multivariate Behav Res* 2011; 46: 399–424.

20. Austin PC. Balance diagnostics for comparing the distribution of baseline covariates between treatment groups in propensity-score matched samples. *Stat Med* 2009; 28: 3083–3107.

21. Hassi Roman M, Mate K, De Pablos-Rodriguez P, et al. Prostate-specific antigen at 3 months as a predictor of radiologic progression-free survival in metastatic hormone-sensitive prostate cancer treated with apalutamide: analysis of 633 patients in a real-world database. *Eur Urol Open Sci* 2024; 70: 58–63.

22. Lopez-Abad A, Belmonte M, Ramirez Backhaus M, et al. Ultralow prostate-specific antigen (PSA) levels and improved oncological outcomes in metastatic hormone-sensitive prostate cancer (mHSPC) patients treated with apalutamide: a real-world multicentre study. *J Clin Med* 2024; 13: 20241018.

23. Maughan BL, Y, Mundle S, Wang X, et al. Rapid and deep prostate-specific antigen (PSA) response to apalutamide plus ADT correlates with improved survival in metastatic castration-sensitive prostate cancer (mCSPC) in real world practice in the US (OASIS Project). In: *2024 American society of clinical oncology annual meeting* Chicago, IL, USA, 2024.

24. Miszczyk M, Fazekas T, Rajwa P, et al. Prostate-specific antigen response as a prognostic factor for overall survival in patients with prostate cancer treated with androgen receptor pathway inhibitors: a systematic review and meta-analysis. *Eur Urol Focus* 2025; 11(5): 755–766.

25. Santoni M, Buttner T, Rescigno P, et al. Apalutamide in metastatic castration-sensitive prostate cancer: results from the multicenter real-world ARON-3 study. *Eur Urol Oncol* 2025; 8: 444–451.

26. Roy S, Sun Y, Chi KN, et al. Early prostate-specific antigen response by 6 months is predictive of treatment effect in metastatic hormone sensitive prostate cancer: an exploratory analysis of the TITAN trial. *J Urol* 2024; 212: 672–681.

27. Bilen MA, Lowentritt B, Khilfeh I, et al. Overall survival with apalutamide versus enzalutamide in metastatic castration-sensitive prostate cancer. *Adv Ther* 2025; 42: 3437–3454.

28. Small EJ, Chi KN, Chowdhury S, et al. Post hoc analysis of rapid and deep prostate-specific antigen decline and patient-reported health-related quality of life in SPARTAN and TITAN patients with advanced prostate cancer. *Eur Urol Oncol* 2024; 7: 844–852.

Visit Sage journals online journals.sagepub.com/home/tam

Sage journals